

Monopsony and the Misallocation of Labour in Indian Organised Manufacturing: A Hsieh–Klenow Decomposition, 2012–13 to 2023–24

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Abstract

This paper measures the misallocation of labour across plants in Indian organised manufacturing and traces it to its source. Using Annual Survey of Industries (ASI) unit-level data for 2012–13 to 2023–24 ($N = 478,442$ plant-year observations across 22 two-digit NIC industries) and the HsiehKlenow2009 accounting framework, we document substantial dispersion in the marginal revenue product of labour (MRPL): within industry-year cells the standard deviation of log MRPL is 1.20, against 0.36 for revenue productivity (log TFPR). Equalising marginal revenue products within industries would raise manufacturing TFP by +102% in the canonical value-added implementation at an elasticity of substitution $\sigma = 3$ (in the 100–128% range HsiehKlenow2009 report for India), and by a conservative +16% in an integrated gross-output implementation. Dispersion has fallen modestly over the period. Our central contribution is to decompose labour misallocation using the exact identity

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MRPL = $w \times \psi \times \mu^M$, where ψ is the plant’s labour-market markdown (monopsony power) and μ^M its product-market markup, both estimated in the companion paper (Objective 1). **The markdown alone accounts for 78.8% of the variance of log MRPL**; the net market-power channel accounts for 64.6%; equalising the markdown within industries would cut the standard deviation of log MRPL from 1.20 to 0.80 (a 55.7% variance reduction); and the cross-industry correlation between labour misallocation and markdown dispersion is 0.83. The decomposition is robust to the winsorisation threshold, sub-periods, plant-size and balanced-panel restrictions, the cell definition, and the assumed rental rate. Labour misallocation in Indian manufacturing is, first and foremost, a monopsony phenomenon—implying that competition and labour-market policy carry first-order allocative-efficiency gains, not merely distributional ones.

JEL codes: D24, D61, J42, L11, O14, O47.

Keywords: misallocation, monopsony, markdown, total factor productivity, Hsieh–Klenow, marginal revenue product, Indian manufacturing, ASI.

1 Introduction

A central question in development macroeconomics is why total factor productivity (TFP) is so much lower in poor countries. HsiehKlenow2009 provided an influential answer: a large part of the gap reflects not differences in technology but the *misallocation* of inputs across heterogeneous plants. When marginal revenue products are equated across plants within an industry, output is maximised for given technology and factor supplies; when they are not, resources are trapped in low-value uses and aggregate TFP falls below its efficient frontier. Hsieh and Klenow estimated that equalising marginal revenue products within four-digit industries would raise manufacturing TFP by 100–128% in India and 30–50% in China relative to the United States.

A decade and a half of subsequent work has measured the *size* of misallocation in many settings, but has been less successful at identifying its *sources*. Candidate frictions—credit constraints, size-dependent regulation, adjustment costs, and distortionary taxation—are

numerous, and the HsiehKlenow2009 accounting is, by design, agnostic about which wedge is responsible. This paper makes progress on the sources question for one factor—labour—by exploiting a tight link between misallocation and *market power*. The marginal revenue product of labour can be written exactly as the wage times two estimable wedges: the labour-market markdown (monopsony power) and the product-market markup. This lets us ask, and answer, how much of the observed dispersion in the marginal revenue product of labour is attributable to monopsony.

We study Indian organised manufacturing over 2012–13 to 2023–24 using plant-level Annual Survey of Industries (ASI) data. The analysis builds directly on the companion paper (Objective 1 of this dissertation), which estimated, for the same 478,442-plant-year panel, LP+ACF Cobb–Douglas production functions, De Loecker–Warzynski product markups μ^M , and BrooksKaboskiLi2021 labour markdowns $\psi = \mu^L/\mu^M$. Because the marginal revenue product of labour equals the labour markup times the wage, $\text{MRPL} = \mu^L w$, and $\mu^L = \psi \mu^M$, the marginal revenue product factors as

$$\text{MRPL} = \underbrace{w}_{\text{wage}} \times \underbrace{\psi}_{\text{markdown (monopsony)}} \times \underbrace{\mu^M}_{\text{markup (product power)}}.$$

Taking logs and removing the industry-year mean—the space in which Hsieh and Klenow measure misallocation—yields an exact variance decomposition of labour misallocation into a wage component and a market-power component. The estimated markdown is, in this sense, *the* Hsieh–Klenow labour wedge, which is what makes the link to market power exact rather than merely suggestive.

Our findings are fourfold. First, labour is substantially mis-priced at the margin: the within-industry-year standard deviation of log MRPL is 1.20 and of log MRPK is 1.45, far exceeding the 0.36 for log TFPR, since the factor wedges partly offset one another in the revenue-productivity index. All measures decline modestly over the sample. Second, the implied output cost is large: +102% in the canonical value-added Hsieh–Klenow implementation at $\sigma = 3$ and +16% in a conservative integrated gross-output implementation, both falling over time. Third, and centrally, labour misallocation is

predominantly a monopsony phenomenon: the markdown accounts for 78.8% of the variance of log MRPL, the net market-power channel for 64.6%, and equalising the markdown would reduce the variance by 55.7%; across industries the correlation between labour misallocation and markdown dispersion is 0.83. Fourth, these results are robust to a battery of checks on trimming, sample, granularity, and parameters.

The paper contributes to three literatures. To the *misallocation* literature HsiehKlenow2009, RestucciaRogerson2008, Banerjee2005, it supplies a direct, data-driven attribution of labour misallocation to a specific, estimable friction—monopsony—rather than to an unexplained residual wedge. To the *monopsony* literature Manning2003, Manning2021, BrooksKaboskiLi2020, it shows that wage-setting power is not only a distributional force depressing the labour share but a quantitatively dominant *allocative* distortion. To the literature on *Indian manufacturing*, it provides the first plant-level misallocation decomposition spanning the GST transition and the COVID-19 shock, and links the falling-markup, persistent-monopsony pattern of Objective 1 to the efficiency of resource allocation.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 presents the misallocation framework, its two implementations, and the market-power decomposition. Section 5 reports the main results. Section 6 presents robustness checks. Section 7 discusses implications. Section 8 concludes.

2 Related Literature

2.1 Misallocation and Aggregate TFP

The modern misallocation literature begins with RestucciaRogerson2008 and HsiehKlenow2009, who formalised how plant-level distortions to the marginal products of inputs lower aggregate TFP even when plant-level technologies are held fixed. In the Hsieh–Klenow framework, plants within an industry face idiosyncratic output and capital “wedges”; the dispersion of revenue productivity (TFPR) is a sufficient statistic for the resulting TFP loss, and the gap between actual and efficient TFP can be sizeable. Banerjee2005 had

earlier argued, from a range of micro evidence, that the marginal product of capital varies enormously across Indian producers, pointing to severe capital misallocation. Subsequent work refined the measurement: Asker2014 show that adjustment costs and capital dynamics inflate measured dispersion if not handled carefully, and that winsorising the tails of the productivity distribution is important; BilsKlenowRuane2021 develop methods to separate genuine misallocation from measurement error using the covariance of inputs and productivity. We adopt the standard tail-winsorisation safeguard and report robustness to it.

2.2 The Sources-of-Misallocation Problem

While the *size* of misallocation is now well documented, attributing it to specific frictions remains difficult, because the Hsieh–Klenow wedges are residuals that absorb everything that makes marginal revenue products differ. RestucciaRogerson2017 survey the literature and emphasise precisely this identification gap. A productive line of work therefore tries to map wedges onto named, measurable distortions—financial frictions, size-dependent policies, contract enforcement, and markups. This paper joins that effort on the labour side by mapping the labour wedge onto an independently estimated measure of employer wage-setting power.

2.3 Market Power as a Source of Misallocation

That markups can generate misallocation is well understood theoretically: heterogeneous markups misallocate resources by driving a wedge between price and marginal cost that varies across producers Edmond2023,Peters2020. Edmond2023 quantify the TFP cost of markup dispersion in a calibrated model and find it substantial. The *labour*-market analogue—heterogeneous markdowns misallocating labour—has received far less empirical attention, despite a large parallel literature establishing that monopsony is pervasive. Manning2003,Manning2021 provide the theory and evidence of dynamic monopsony; SokolovaSorensen2021 meta-analyse over 1,300 estimates and find labour-supply elasticities to the firm around 1.5, implying wages roughly 40% below marginal revenue product;

BergerHerkenhoffMongey2022 embed labour market power in general equilibrium and find large welfare losses. BrooksKaboskiLi2021 supply the production-data measure of the markdown $\psi = \mu^L/\mu^M$ that we use, and apply it to Indian ASI data through 2013. Our contribution is to feed that markdown into the Hsieh–Klenow accounting and show that it is the dominant component of labour misallocation—bridging the markup and monopsony literatures with the misallocation literature.

2.4 Misallocation in Indian Manufacturing

For India specifically, HsiehKlenow2009 remains the benchmark, with its 100–128% potential TFP gain. The persistence of misallocation despite three decades of liberalisation has been linked to size-dependent labour regulation, weak contract enforcement, and the small-scale-industry reservation policy. Our results both corroborate the benchmark magnitude—an external validation of our implementation—and add a new finding: a majority of the labour component of that misallocation is traceable to dispersed monopsony power, a friction that the earlier Indian literature did not measure directly.

3 Data

The data are the plant-level Annual Survey of Industries (ASI) panel constructed in Objective 1, conducted annually by the Ministry of Statistics and Programme Implementation (MOSPI). The ASI covers registered manufacturing establishments under the Factories Act, 1948; each factory is identified by a Dispatch Serial Number (DSL). We use fiscal years 2012–13 to 2023–24 (12 waves). The Objective 1 estimation panel contains 515,374 plant-year observations across 79,704 unique factories.

For the misallocation analysis we apply the sample restrictions of HsiehKlenow2009: we require strictly positive real output, capital, labour, materials, and wage bill, and non-missing, positive estimated output elasticities, and we drop the NIC2 placeholder code (99). This yields a working sample of 478,442 plant-years across 22 two-digit NIC industries. Industry NIC 13 (Textiles) does not survive the joint elasticity restriction and

is absent from the misallocation panel; this is noted throughout. The panel is unbalanced, and all aggregation to the industry and economy level uses the ASI sampling multiplier m_i as a population weight, exactly as in Objective 1.

The key plant-level inputs are inherited unchanged from Objective 1: gross output Y (Block J, WPI-deflated), materials M (Blocks H+I consumed, deflated), labour L (total persons engaged, Block E SNO 9), the wage bill W (emoluments), and capital K (gross fixed assets ex-land, deflated). The plant-level LP+ACF Cobb–Douglas output elasticities $(\theta_s^L, \theta_s^K, \theta_s^M)$, the product markup μ_{it}^M , and the labour markdown ψ_{it} are taken directly from the Objective 1 output panel, so that the misallocation accounting is fully consistent with the market-power estimates.

4 Methodology

4.1 Marginal Revenue Products and Revenue Productivity

For plant i in industry-year cell s with nominal revenue $P_{si}Y_{si}$, the gross-output marginal revenue products of labour and capital are

$$\text{MRPL}_{si} = \theta_s^L \frac{P_{si}Y_{si}}{L_{si}}, \quad \text{MRPK}_{si} = \theta_s^K \frac{P_{si}Y_{si}}{K_{si}}, \quad (1)$$

and revenue productivity (up to an industry-year constant) is

$$\log \text{TFPR}_{si} = \log(P_{si}Y_{si}) - \theta_s^L \log L_{si} - \theta_s^K \log K_{si} - \theta_s^M \log M_{si}. \quad (2)$$

Physical productivity TFPQ replaces nominal with real output; because the two differ only by the plant’s output price, which is common within a cell up to the deflator, *within-cell dispersion* of TFPR and TFPQ is identical. The misallocation measures are the within-cell standard deviations of $\log \text{MRPL}$, $\log \text{MRPK}$, and $\log \text{TFPR}$, where a cell s is a NIC two-digit $\times$ year group.

4.2 Two Implementations

We report two complementary implementations. The **integrated** implementation uses the plant-level gross-output elasticities and revenue from Objective 1 directly in (1)–(2); its labour wedge is, by construction, the estimated markdown, which is what makes the decomposition in Section 4.4 exact. The **canonical** implementation reproduces the original HsiehKlenow2009 value-added Cobb–Douglas setup: value added $VA = Y - M$, an industry capital share $\alpha_s = RK/(RK + wL)$ formed from imputed capital payments with rental rate $R = 0.10$, and the monopolistic-competition wedge $(\sigma - 1)/\sigma$ with elasticity of substitution $\sigma = 3$ (baseline; $\sigma = 4, 5$ as robustness). The canonical version is calibrated to be comparable to Hsieh and Klenow’s own India numbers; the integrated version is the one tied to the market-power estimates.

4.3 Aggregate Efficiency Gain

Following HsiehKlenow2009, define plant physical productivity A_{si} via $\log A_{si} = \log \text{TFPR}_{si} + \frac{1}{\sigma-1} \log(P_{si}Y_{si})$ and the revenue-weighted harmonic mean $\overline{\text{TFPR}}_s$. Efficient and actual industry TFP are

$$\text{TFPR-efficient: } \text{TFP}_s^{\text{eff}} = \left[\sum_i m_i A_{si}^{\sigma-1} \right]^{\frac{1}{\sigma-1}}, \quad \text{TFP}_s = \left[\sum_i m_i (A_{si} \overline{\text{TFPR}}_s / \text{TFPR}_{si})^{\sigma-1} \right]^{\frac{1}{\sigma-1}},$$

the industry gain is $\text{TFP}_s^{\text{eff}}/\text{TFP}_s$, and the economy-wide gain is the value-added-share-weighted product $Y^{\text{eff}}/Y = \prod_s (\text{TFP}_s^{\text{eff}}/\text{TFP}_s)^{\theta_s}$. TFPR and TFPQ are winsorised at the 1st/99th percentiles within each cell before aggregation.

4.4 Decomposing Labour Misallocation into Market Power

Because $\text{MRPL}_{si} = \mu_{si}^L w_{si}$ and the markdown is $\psi_{si} = \mu_{si}^L / \mu_{si}^M$, the marginal revenue product of labour factors exactly as $\text{MRPL}_{si} = w_{si} \psi_{si} \mu_{si}^M$. Removing the industry-year mean (denoted $d \log$) gives the exact identity

$$d \log \text{MRPL}_{si} = d \log w_{si} + d \log \psi_{si} + d \log \mu_{si}^M. \quad (3)$$

196 Labour misallocation— $\text{var}(d \log \text{MRPL})$ within cells—therefore decomposes into a wage
 197 component and a market-power component. The variance shares

$$\text{share}_X = \frac{\text{cov}(d \log \text{MRPL}, d \log X)}{\text{var}(d \log \text{MRPL})}, \quad X \in \{w, \psi, \mu^M\}, \quad (4)$$

198 sum to one. We complement the shares with counterfactuals that equalise a wedge within
 199 each cell (set its deviation to zero) and recompute the dispersion of $\log \text{MRPL}$, holding
 200 physical productivity fixed. Crucially, this attribution operates entirely within the observed
 201 MRPL distribution and requires no value of σ .

202 5 Results

203 5.1 The Extent of Misallocation

204 Pooling all plant-years and measuring deviations within $\text{NIC2} \times \text{year}$ cells, the standard
 205 deviation of $\log \text{MRPL}$ is 1.20, of $\log \text{MRPK}$ is 1.45, and of $\log \text{TFPR}$ is 0.36 (Table 1).
 206 The dispersion of the estimated markdown $\log \psi$ is 1.22—almost identical to that of MRPL ,
 207 a first sign that labour misallocation is tied to monopsony. That the marginal revenue
 208 products are far more dispersed than TFPR is expected, since the labour and capital
 209 wedges partly offset one another in the revenue-productivity index. The 90–10 $\log$ gap
 210 in MRPL is large and stable (around 3.0, i.e. a factor of ≈ 20 between the 90th- and
 211 10th-percentile plants within a cell), while the corresponding TFPR gap is far smaller
 212 (≈ 0.7 – 0.9). Our TFPR dispersion (0.36) is below the ≈ 0.67 HsiehKlenow2009 report
 213 for India, reflecting our gross-output revenue productivity, within-cell winsorisation, and
 214 removal of industry-year effects.

215 **Evolution, 2012–24.** Misallocation has *declined modestly*. The within-industry disper-
 216 sion of TFPR falls from 0.410 in 2012–13 to 0.346 in 2023–24 (a 16% reduction in the
 217 standard deviation), with MRPL dispersion falling from 1.271 to 1.179 and MRPK disper-
 218 sion from 1.551 to 1.346. The markdown dispersion is essentially flat ($1.251 \rightarrow 1.202$). The
 219 improvement is gradual and broad-based rather than driven by any single year (Figure 1).

Table 1: Within-industry-year dispersion of marginal revenue products and TFPR

Year	SD log MRPL	SD log MRPK	SD log TFPR	SD log ψ	SD log TFPR ^{HK}
2012–13	1.271	1.551	0.410	1.251	1.000
2013–14	1.230	1.549	0.385	1.232	0.971
2015–16	1.222	1.604	0.376	1.246	0.958
2017–18	1.200	1.447	0.360	1.220	0.925
2019–20	1.168	1.384	0.354	1.207	0.902
2021–22	1.182	1.344	0.336	1.199	0.906
2023–24	1.179	1.346	0.346	1.202	0.902
Pooled	1.202	1.451	0.363	1.221	—

Note: Pooled deviations within NIC2×year cells; TFPR winsorised 1/99 within cell. TFPR^{HK} is the canonical value-added Hsieh–Klenow revenue productivity (robustness).

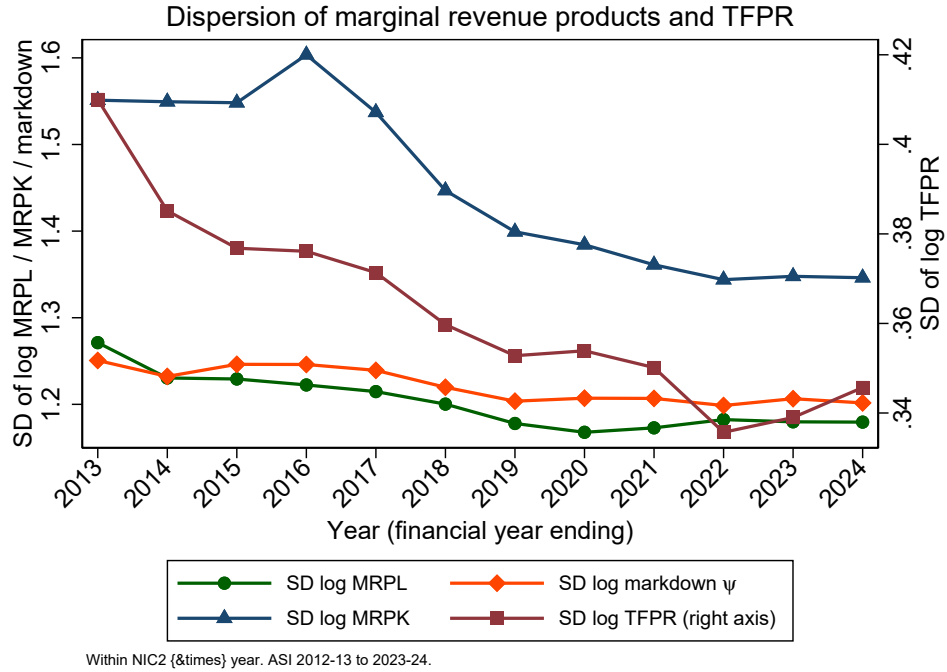


Figure 1: Within-industry-year dispersion of log MRPL, log MRPK, log TFPR and log ψ , 2012–13 to 2023–24. Source: ASI; authors’ calculations.

5.2 The Cost of Misallocation

Table 2 reports the aggregate efficiency gain—the percentage rise in manufacturing TFP from equalising marginal revenue products within industries (multiplier-weighted; Figure 2). In the canonical Hsieh–Klenow value-added framework the gain is +102% at $\sigma = 3$, rising to +139% and +169% at $\sigma = 4$ and $\sigma = 5$. This is squarely within the 100–128% range HsiehKlenow2009 report for India, an important external validation. The integrated gross-output measure, which treats materials as a freely-adjusted input and is therefore a conservative lower bound, gives +16% at $\sigma = 3$ (+23% and +32% at $\sigma = 4, 5$). Under both measures the potential gain *falls* over the period (canonical 136% $\rightarrow$ 92%; integrated 23% $\rightarrow$ 13%), consistent with the declining dispersion.

Table 2: Aggregate efficiency gain $Y^{\text{eff}}/Y - 1$ (%), pooled 2012–24

Approach	$\sigma = 3$	$\sigma = 4$	$\sigma = 5$
Canonical Hsieh–Klenow (value added)	102.2	138.9	169.3
Integrated (gross output)	15.6	23.1	32.3

Note: Multiplier-weighted; TFPR/TFPQ winsorised 1/99 within NIC2 $\times$ year. 264 industry-year cells (22 industries $\times$ 12 years).

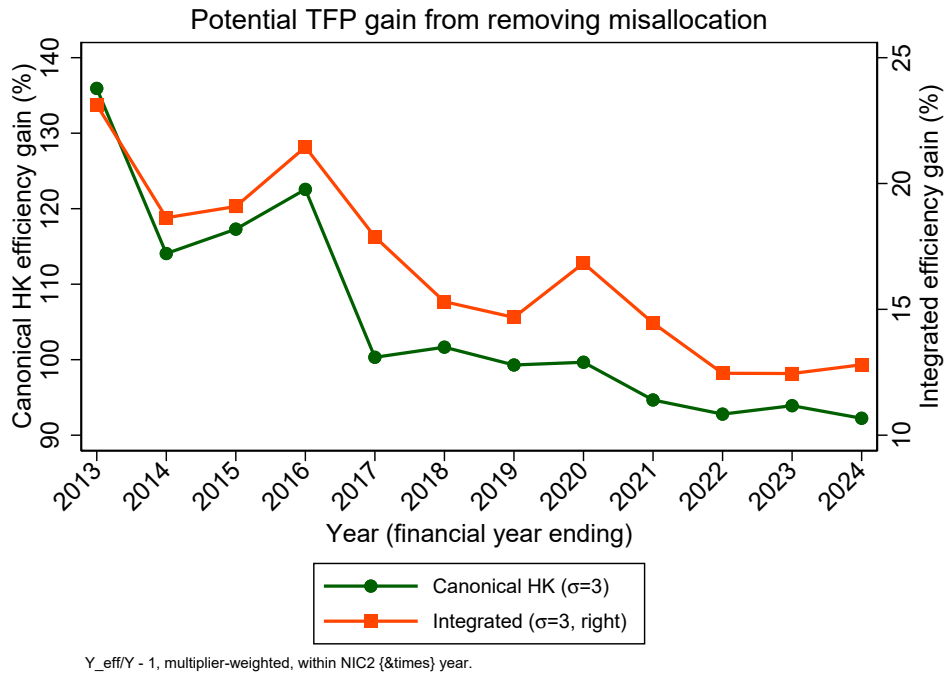


Figure 2: Aggregate efficiency gain over time, canonical and integrated implementations ($\sigma = 3$). Source: ASI; authors’ calculations.

5.3 Heterogeneity Across Industries and States

Misallocation is highly heterogeneous across industries (Table 3; Figure 3). TFPR dispersion is greatest in Beverages (NIC 11, 0.675), Printing (NIC 18, 0.595), Pharmaceuticals (NIC 21, 0.511), Other non-metallic minerals (NIC 23, 0.476), Tobacco (NIC 12, 0.437) and Computers/electronics (NIC 26, 0.431); it is lowest in Paper (NIC 17, 0.227), Rubber and plastics (NIC 22, 0.245), Basic metals (NIC 24, 0.277) and Motor vehicles (NIC 29, 0.287). Labour misallocation (SD log MRPL) is largest in Tobacco (1.82), Coke and refined petroleum (NIC 19, 1.64) and Other non-metallic minerals (1.49).

Table 3: Industries with the highest and lowest TFPR dispersion

	Industry (NIC-2008)	SD log TFPR	SD log MRPL
Highest	11 Beverages	0.675	1.346
	18 Printing	0.595	1.106
	21 Pharmaceuticals	0.511	1.062
Lowest	17 Paper	0.227	0.975
	22 Rubber & plastics	0.245	0.959
	24 Basic metals	0.277	1.227

Note: Within-cell SD pooled over years. The complete 22-industry table is reported in Appendix A.

Across the large manufacturing states, within-industry TFPR dispersion clusters in a narrow band: Maharashtra 0.368, Karnataka 0.373, Tamil Nadu 0.358, Gujarat 0.347, Uttar Pradesh 0.333. Labour (MRPL) dispersion is somewhat higher in the eastern and central states (Jharkhand 1.35, Bihar 1.32, Rajasthan 1.20) than in the southern industrial states. The full state table, with Census-2011 state names attached, is available from the authors.

5.4 Labour Misallocation Is a Monopsony Phenomenon

We now turn to the central result. Identity (3) holds in the data up to the small slippage induced by the winsorisation of the markdown in Objective 1 (median residual ≈ 0.005 ; residual variance $\approx 3\%$ of $\text{var}(d \log \text{MRPL})$). Table 4 reports the variance shares (4). **The markdown accounts for 78.8% of the variance of log MRPL**, wage dispersion for 29.8%, and the product markup for -14.2% . The markup share is negative because,

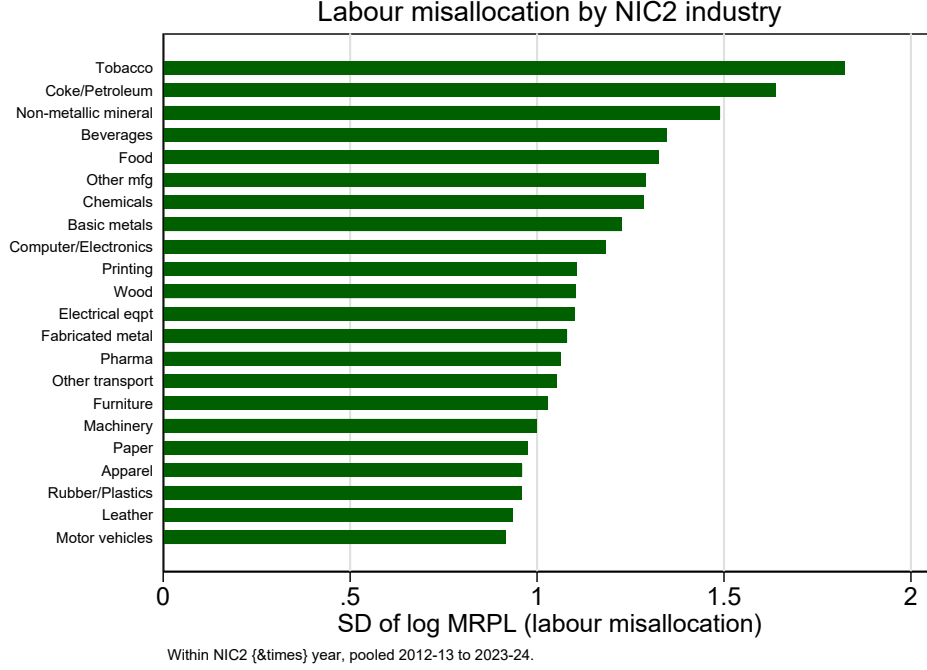


Figure 3: Labour misallocation (SD log MRPL) and revenue-productivity dispersion (SD log TFPR) by NIC2 industry. Source: ASI; authors' calculations.

by $\psi = \mu^L/\mu^M$, a higher markup mechanically lowers the markdown, so μ^M covaries negatively with MRPL at the plant level. The *net* market-power channel (markdown plus markup) is 64.6%. Labour misallocation in Indian organised manufacturing is thus predominantly a monopsony phenomenon.

Table 4: Variance-share decomposition of labour misallocation, $\text{var}(d \log \text{MRPL}) = 1.44$

Component	Share of $\text{var}(\log \text{MRPL})$
Wage dispersion	0.298
Markdown ψ (monopsony)	0.788
Product markup μ^M	-0.142
Market power ($\psi + \mu^M$)	0.646

Note: Shares from $\text{cov}(d \log \text{MRPL}, d \log X) / \text{var}(d \log \text{MRPL})$ and sum to one.

Counterfactuals. Table 5 reports the dispersion of log MRPL under hypothetical equalisation of the wedges within each industry, holding physical productivity fixed. **Equalising the markdown alone cuts the variance of log MRPL by 55.7%** (the standard deviation falls from 1.202 to 0.800). Equalising both market-power wedges leaves only wage

dispersion and reduces the variance by 68.7% (SD = 0.672). Removing the markup *in isolation* raises MRPL dispersion (variance +39%), confirming that the markup partly offsets the markdown at the plant level.

Table 5: Counterfactual dispersion of log MRPL under wedge equalisation

Scenario	SD log MRPL	Variance reduction
Actual	1.202	—
Equalise markdown ψ	0.800	−55.7%
Equalise markup μ^M	1.418	+39.2%
Equalise both	0.672	−68.7%

Stability and the cross-industry pattern. The decomposition is remarkably stable over time (Figure 4): the markdown share is 0.75 in 2012–13 and rises slightly to 0.81 by 2022–23, while the market-power share holds near 0.65 and the markdown-equalisation reduction stays in the 55–58% range. Across industries the relationship is strong: the correlation between an industry’s labour misallocation (SD log MRPL) and its markdown dispersion (SD log ψ) is 0.83, against 0.43 for markup dispersion. Industries with more heterogeneous monopsony power are systematically more labour-misallocated.

6 Robustness

We subject the two headline results—the dispersion magnitudes and the monopsony attribution—to four families of robustness checks, all run on the real panel.

R1: Winsorisation. Table 6 recomputes the pooled within-cell dispersion under no trimming and under 1/99, 2/98 and 5/95 trimming applied to each measure. The ordering is preserved and the magnitudes change only gradually: SD log MRPL moves from 1.20 (raw) to 1.07 (5/95), and SD log TFPR from 0.42 to 0.30. The markdown dispersion is essentially invariant. The dispersion is not an artefact of tail observations.

R2: Sub-periods and sample restrictions. Table 7 re-estimates the dispersion and the monopsony attribution on seven subsamples. The markdown share of labour

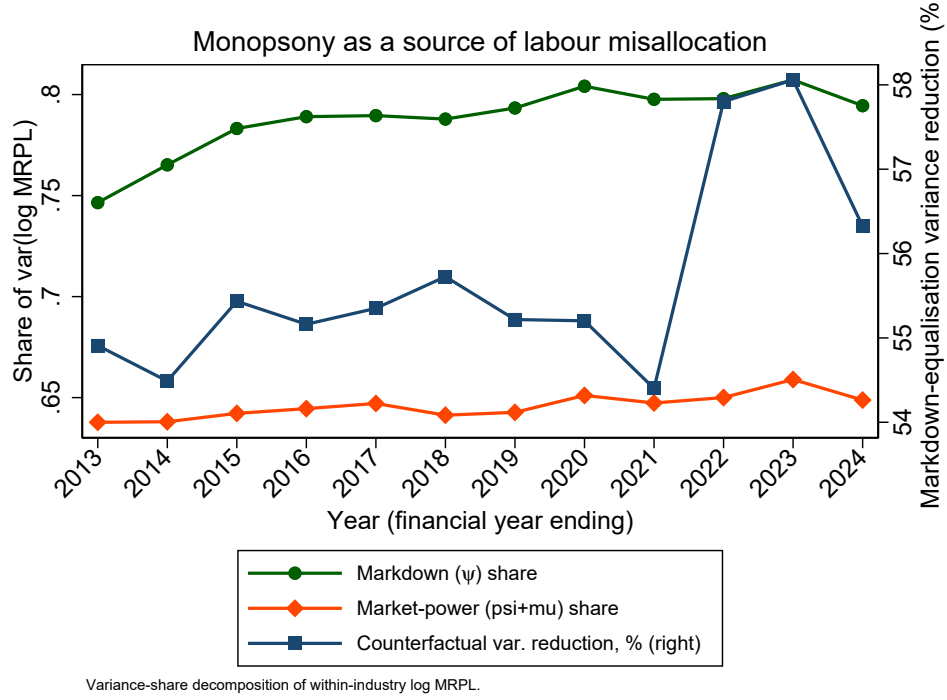


Figure 4: Markdown share of labour misallocation and markdown-equalisation variance reduction, by year. Source: ASI; authors' calculations.

Table 6: R1 — Dispersion by winsorisation threshold (pooled, within NIC2×year)

Trim	SD log MRPL	SD log MRPK	SD log TFPR	SD log ψ
Raw (none)	1.202	1.451	0.416	1.221
1/99	1.167	1.307	0.363	1.221
2/98	1.141	1.252	0.340	1.221
5/95	1.074	1.138	0.299	1.146

misallocation lies between 0.77 and 0.84 throughout—pre-GST and post-GST, balanced panel, after dropping the tobacco (12) and coke/petroleum (19) outliers, and for large (≥ 100 persons) and small plants alike—and the markdown-equalisation reduction stays in the 49–59% band. The “monopsony dominates” conclusion is not an artefact of any particular period or plant type.

Table 7: R2 — Dispersion and monopsony attribution by subsample

Subsample	N	SD log MRPL	SD log TFPR	Markdown share	Mktpwr share	ψ
Full sample	478,442	1.202	0.363	0.788	0.646	
Pre-GST (2013–17)	190,685	1.233	0.384	0.775	0.642	
Post-GST (2018–24)	287,757	1.180	0.348	0.798	0.649	
Balanced panel	4,848	1.050	0.359	0.840	0.683	
Drop NIC 12 & 19	465,705	1.184	0.362	0.795	0.650	
Large (≥ 100)	202,098	1.110	0.366	0.773	0.637	
Small (< 100)	276,344	1.231	0.359	0.829	0.682	

Note: “Markdown share” and “Mktpwr share” are the variance shares of log MRPL from (4); the final column reduction from equalising the markdown within cells.

R3: Rental rate. The canonical capital share $\alpha_s = RK/(RK + wL)$ depends on the assumed rental rate R . Table 8 recomputes the pooled efficiency gain ($\sigma = 3$) for $R \in \{0.08, 0.10, 0.12\}$. The gain is essentially flat—102.2%, 102.1% and 103.2% respectively—so the headline cost of misallocation is insensitive to R . (Only the elasticity σ scales the gain materially; see Table 2.)

Table 8: R3 — Canonical efficiency gain ($\sigma = 3$) by rental rate R

Rental rate R	0.08	0.10	0.12
Efficiency gain $Y^{\text{eff}}/Y - 1$ (%)	102.2	102.1	103.2

R3b: Cell definition. Table 9 re-measures dispersion and the markdown share when “within industry” is defined at NIC2×year, NIC3×year, or NIC2×state×year (cells with fewer than 5 plants dropped). The markdown share ranges only from 0.77 to 0.79, and the market-power share from 0.64 to 0.67; the result is robust to how finely the misallocation cell is drawn.

Table 9: R3b — Dispersion and markdown share by cell definition

Cell definition	N	SD log MRPL	SD log TFPR	Markdown share	Mktpwr share
NIC2 $\times$ year	478,442	1.202	0.363	0.788	0.646
NIC3 $\times$ year	478,351	1.166	0.361	0.770	0.636
NIC2 $\times$ state $\times$ year	475,486	1.110	0.347	0.789	0.674

7 Discussion

The results reframe how we should think about both misallocation and monopsony in Indian manufacturing. The misallocation literature has long treated the labour wedge as an unexplained residual; we show that, for India, a majority of its variance is explained by a single, independently measured friction—employer wage-setting power. The monopsony literature, in turn, has emphasised the distributional consequences of markdowns (a depressed labour share, documented in Objective 1); we show that the same markdowns are the dominant *allocative* distortion in the labour market, with the dispersion they generate mapping into a first-order TFP loss.

Three implications follow. First, the efficiency case for competition policy extends to the labour market. Because monopsony—not idiosyncratic wage noise or product-market power—is the principal source of labour misallocation, policies that curb employer wage-setting power (minimum-wage enforcement, benefit portability, lower mobility frictions, scrutiny of employer concentration) would yield allocative gains, not merely redistribution. Second, product- and labour-market policy are complements: the markup partly offsets the markdown at the plant level, so addressing one wedge in isolation can be counterproductive, whereas the largest gains come from compressing the markdown. Third, the modest decline in dispersion over 2012–24 suggests that the deepening product-market competition documented in Objective 1 (the falling markup) has been accompanied by only a gradual easing of labour-market distortions—monopsony has proved more persistent than market power.

The analysis has limitations. The technology is Cobb–Douglas (Translog was rejected on the short ASI panel), so elasticities are industry-constant; the markdown was winsorised, slightly attenuating the tails (residual variance $\approx 3\%$); and the accounting is descriptive—it

apportions the variance of MRPL but does not, alone, identify the structural origins of monopsony (search frictions, spatial concentration, employer concentration). That last point motivates the next chapter, which links the markdown dispersion measured here to labour-market concentration (the state $\times$ NIC employment HHI) and to the macroeconomic and spatial determinants of market power.

8 Conclusion

Using the Hsieh–Klenow framework on a 478,442-plant-year ASI panel, this paper has shown that labour is substantially mis-priced at the margin in Indian organised manufacturing (SD log MRPL = 1.20), that equalising marginal revenue products within industries would raise manufacturing TFP by roughly +102% (canonical) or +16% (integrated lower bound), and—its central contribution—that this labour misallocation is predominantly a monopsony phenomenon: the estimated markdown accounts for 78.8% of the variance of log MRPL, equalising it would cut the dispersion by 55.7%, and across industries the correlation between labour misallocation and markdown dispersion is 0.83. The finding is robust to winsorisation, sub-periods, plant size, cell granularity, and the rental rate. The market power documented in Objective 1 is therefore not merely a distributional phenomenon but the quantitatively dominant source of the misallocation of labour across Indian manufacturing plants—the empirical core of this dissertation’s integration of market power and misallocation, and the point of departure for the determinants analysis of Chapter 6.

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A Full Dispersion by NIC2 Industry

Table 10: Within-industry-year dispersion by NIC2 industry (pooled 2012–24)

NIC2	Industry (NIC-2008)	N	SD log MRPL	SD log MRPK	SD log TFPR	SD log ψ
10	Food products	78,032	1.326	1.643	0.299	1.308
11	Beverages	9,715	1.346	1.619	0.675	1.651
12	Tobacco products	6,604	1.822	2.440	0.437	1.482
14	Wearing apparel	24,047	0.960	1.243	0.334	0.963
15	Leather & products	10,371	0.935	1.190	0.291	0.916
16	Wood & products	11,296	1.104	1.582	0.259	1.070
17	Paper & products	15,536	0.975	1.131	0.227	0.856
18	Printing	8,287	1.106	1.352	0.595	1.040
19	Coke & petroleum	6,133	1.638	1.856	0.344	1.367
20	Chemicals	34,571	1.286	1.442	0.416	1.200
21	Pharmaceuticals	17,902	1.062	1.344	0.511	1.170
22	Rubber & plastics	29,705	0.959	1.244	0.245	0.971
23	Non-metallic minerals	51,941	1.490	1.550	0.476	1.673
24	Basic metals	30,578	1.227	1.441	0.277	1.292
25	Fabricated metals	29,252	1.080	1.350	0.307	1.092
26	Electronics & optical	9,797	1.184	1.468	0.431	1.322
27	Electrical equipment	26,385	1.100	1.334	0.291	1.050
28	Machinery & equipment	32,692	1.000	1.273	0.316	0.996
29	Motor vehicles	21,416	0.916	1.228	0.287	0.872
30	Other transport	8,859	1.053	1.340	0.351	1.065
31	Furniture	6,737	1.029	1.410	0.287	1.038
32	Other manufacturing	8,586	1.290	1.576	0.413	1.419

Note: Standard deviation of each log measure, computed on deviations from the NIC2×year mean and pooled over 2012–24; log ψ is the estimated labour-market markdown. Industry NIC 13 (Textiles) does not survive the joint elasticity restriction and is absent from the panel.

B Declarations

Data availability. The study uses unit-level Annual Survey of Industries (ASI) microdata of the Ministry of Statistics and Programme Implementation (MoSPI), Government of India. The microdata were obtained under MoSPI’s data-access terms and cannot be redistributed by the authors; they are available to bona fide researchers directly from MoSPI. The derived estimates reported in the paper and the estimation code are available from the corresponding author on reasonable request.

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382 Both authors read and approved the final version.